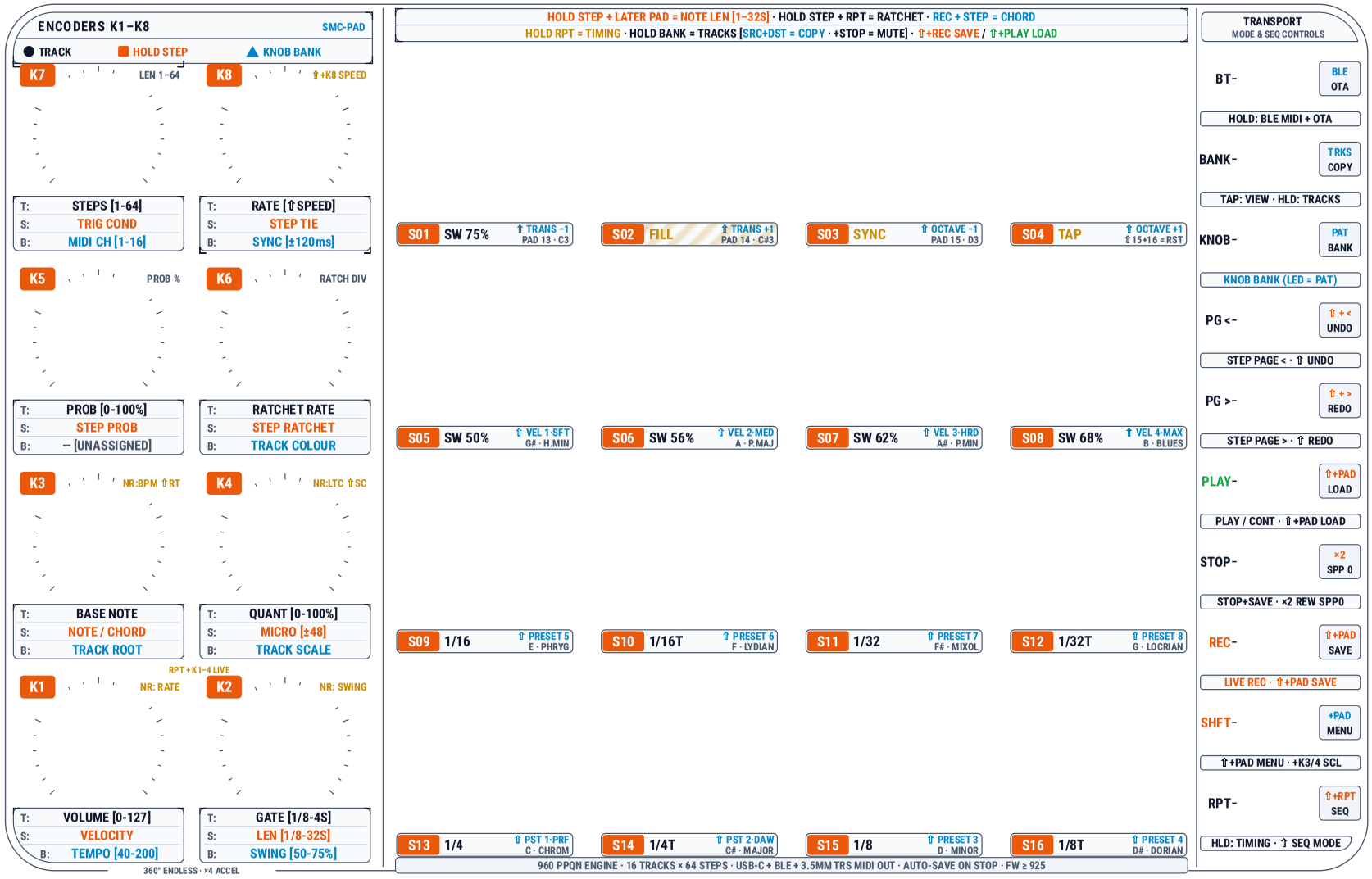


mark distance from document: Left=15.0mm, Top=15.0mm; mark to mark distance: X=185.9mm, Y=230.0mm;

SMC-PAD // CAMEO SHEET 1 OF 2: TOP CONTROL DECK (US LETTER 8.5×11" · PRINT 100% · INKSCAPE X-OFFSET = 6 MM)

SCALE CHECK: 100.00 MM VERNIER



360° PINWHEEL WRAP:

1. Top Deck (flush inside rim)
2. Rear → Left (butt at SEAM)
3. Front → Right (butt at SEAM)
4. I/O Insert (inside rear well)

1:1 SCALE · US LETTER 8.5×11

CNR-FR — FRONT SEAM — STOP×2=0 — R10.7 · 90°

RIGHT-01 // ROOTS & SCALES [P1–12]

P01: C · Chromatic	P07: F# · Mixolydian
P02: C# · Major	P08: G · Locrian
P03: D · Nat Minor	P09: G# · Harm Minor
P04: D# · Dorian	P10: A · Pent Major
P05: E · Phrygian	P11: A# · Pent Minor
P06: F · Lydian	P12: B · Blues

0 LIT = FOLLOW · BANK K3/4 = TRK · ⇧+K3/4 = GLOBAL

RIGHT-02 // LED READOUTS & CLOCK

BAR LEDS:	Vol / Vel / Prob / Qnt / BPM: P1→16
SIGNED:	Micro & Sync: P8..1 (−) · P9..16 (+)
NOTE KEY:	P1–12 = C..B Root · P13–16 = Octave
EXT SYNC:	P15 On = 24 PPQN · STOP×2 = SPP 0
SYNC MS:	BANK K8: −120..+120ms (Master/Slave)

POWER · CHG · USB-C · MIDI

CNR-BL — REAR SEAM — SEAM ——► — R10.7 · 90°

LEFT-01 // K1–K8 KNOB MATRIX

K1	Vol 0–127	Velocity	BPM 40–200
K2	Gate 1/8–4	Len 1/8–32	Swing 50–75
K3	Base Note	Note/Chord	Trk Root
K4	Quantize	Micro ±48	Trk Scale
K5	Prob %	Step Prob	—
K6	Ratch Rate	Stp Ratch	Trk Colour
K7	Steps 1–64	Trig Cond	MIDI Ch
K8	Grid Rate	Step Tie	Sync ±120ms

LEFT-02 // TRIG, GATE & RATCHET

TRIG 1–6:	P1:ALW · P2-3: 1:2, 2:2 · P4-6: 1..3:3
TRIG 7–13:	P7-10: 1..4:4 · P11:1ST · P12:FILL · 13:!F
GATE [K2]:	1/8..7/8, 1..4S · Step Len: 1..32S
RATCH [K6]:	AUTO(split) · 1/8 · 1/16 · 16T · 32 · 64
GRID [K8]:	Keeps timing (dim=off-grid) · ⇧+K8 Speed

POWER · CHG · USB-C · MIDI

CNR-FL — KNOB SIDE — ⇧+RPT=SEQ — FW ≥ 925

F1 // STEP & NOTE LENGTH

TAP PAD:	Step On / Off (S01..S16)
HOLD+PAD:	Note Length 1–32 steps!
HOLD+KNOB:	Step P-Lock (K3 moves chord)
HOLD+RPT:	Ratchet ×1 → ×2 → ×3 → ×4
PG < / >:	Page 1–4 · ⇧+< / > Undo/Redo

F2 // TRACKS [HOLD BANK]

TAP BANK:	Step View ↔ Play View
HOLD+PAD:	Select active Track 1–16
SRC+DST:	Hold Src+Tap Dst = Copy Notes
+⇧ COPY:	Full Copy (+Ch / Note / Vol)
+STOP+PAD:	Mute Trk · +⇧+STOP = Clear

F3 // LIVE REC & TIMING

REC+PLAY:	Live Rec (hold = gate len)
CHORDS:	Hit active step (up to 4 notes)
QUANTIZE:	K4 (0–100%) non-destructive!
HOLD RPT:	P1-8 Rate · P9-13 Swg · P14 Fill
SYNC/TAP:	P15 Ext Clock · P16 Tap Tempo

F4 // 16 PATTERNS & CUE

⇧+REC+PAD:	Save Slot 1–16 (no pad=cur)
⇧+PLAY+PAD:	Load Slot (cues next bar!)
SLOT LEDS:	Off=Empty · Dim=Used · Lit=Cur
FEEDBACK:	Blinks 3× GRN=OK / RED=ERR
AUTO-SAVE:	STOP saves edits to flash

CNR-BR — BTN SIDE — SEAM ——► — R10.7 · 90°

REAR-01 // HARDWARE SCHEMATIC [JIELI AC636N · BR25]

- 16× VEL PADS — MUX PB2/3 → PA4/6
- 8× ENC + 10× BTN — SPI2 595 → PB0/PC1
- 3.7V LI-ION BAT — 704060 · 2000mAh

→ AC636N · BR25 — pi32v2 · 960 PPQN — PI32V2 · 512K NOR — RAM ISR SAFE

- 16× WS2812 RGB — SPI1 PB1 · 800kHz
- TRS MIDI OUT — UART1 PC5 · 31.25k
- USB-C + BLE MIDI — 3-CBL · AE40 OTA

REAR-02 // FLASH MAP, SYSEX & SAFE-MODE RECOVERY

FLASH 512K:	0x00000 CODE · 0x30000 16×PAT · 0x50000 VM · 0x77000 PST
USB 3-CBL:	CBL0: Master MIDI · CBL1: Private Sinco · CBL2: Mackie
DIAG SYSEX:	F0 7D 53 50 01/02/05 F7 (status / counters / crash dump)
SAFE MODE:	Power-cycle 2× in 5s → skips pads/keys, keeps BLE + OTA!
UBOOT ROM:	USB_KEY handshake on DP/DM enters BR25 Mask-ROM loader

PWR · CHG · USB-C · MIDI OUT

5V ⎓ 500mA · RED=CHG / GRN=FULL · TRS-A MIDI OUT

mark distance from document: Left=15.0mm, Top=15.0mm; mark to mark distance: X=185.9mm, Y=230.0mm;

SMC-PAD // CAMEO SHEET 2 OF 2: 360° PINWHEEL SIDES + 2× I/O INSERTS (US LETTER · PRINT 100% · X-OFFSET = 6 MM)